

DATA ANALYTICS & BUSINESS INTELLIGENCE

Road Accident Dashboard

*An interactive Microsoft Excel dashboard built to turn
five years of raw accident records into clear, decision-
ready insight.*

Project Report

Prepared June 2026

Tools: Microsoft Excel · Pivot Tables · Pivot Charts · Slicers

1. Executive Summary	3
2. Project Overview & Objectives	3
3. Tools & Dashboard Features	3
4. Walkthrough of the Dashboard Views	4
5. Chart Reference Guide	6
6. Key Insights & Recommendations	7
7. Closing Thoughts	7

1.Executive Summary

Five thousand accidents, one dashboard, and a much clearer picture of where the risk lives.

This report walks through a road accident analytics dashboard built in Microsoft Excel, covering accident records from 2023 and 2024 across ten Indian cities. Rather than leaving the numbers buried in a spreadsheet, the dashboard organizes them into KPI cards, charts, and interactive filters so that a traffic analyst, city planner, or safety officer can ask a question — "which roads are riskiest in foggy weather?" — and get an answer in seconds, not hours.

The headline numbers are sobering: **5,000 accidents**, **12,700 casualties**, and nearly **14,800 vehicles** involved across the two-year period. Urban roads carry the largest share of incidents, Bhopal stands out as the most consistently accident-prone city, and weather conditions like fog noticeably shift both where and how severely accidents occur. The sections below unpack how the dashboard reaches these conclusions and what they suggest for action.

2.Project Overview & Objectives

The Road Accident Dashboard was built to make a large, messy accident dataset usable. Raw rows of data rarely tell a story on their own — it takes structure, visualization, and the ability to slice the data from different angles before patterns start to emerge. That's exactly what this dashboard does: it takes accident records and reorganizes them into a format where trends across cities, months, weather conditions, and road types become visible at a glance.

The project was guided by a few clear goals:

- Give a real-time, at-a-glance view of overall accident statistics
- Surface monthly and seasonal trends rather than just year-end totals
- Track casualties and vehicle involvement alongside raw accident counts, since frequency alone doesn't capture severity
- Pinpoint the cities and road types that carry the most risk
- Make weather-related risk visible, since conditions like fog and rain are largely predictable and preventable
- Let users explore freely through slicers, without ever touching or risking the underlying data

3.Tools & Dashboard Features

Everything here was built natively in Excel — no external BI software — using Pivot Tables and Pivot Charts as the analytical backbone, Slicers for interactivity, KPI cards for quick reads, and Conditional Formatting to draw the eye to what matters. The data itself went through a cleaning pass before analysis to handle inconsistencies and missing values.

Interactive Filters

Four slicers sit on the right-hand side of the dashboard — **Year** (2023 & 2024), **Weather Condition**, **Road Type**, and **City** — and they can be combined freely. Selecting any combination instantly refreshes every KPI card and chart on the page.

KPI Cards

Three cards anchor the top of the dashboard: **Total Accidents**, **Total Casualties**, and **Vehicles Involved**. They're designed to answer the first question anyone asks — "how bad is it?" — before a single chart is even read.

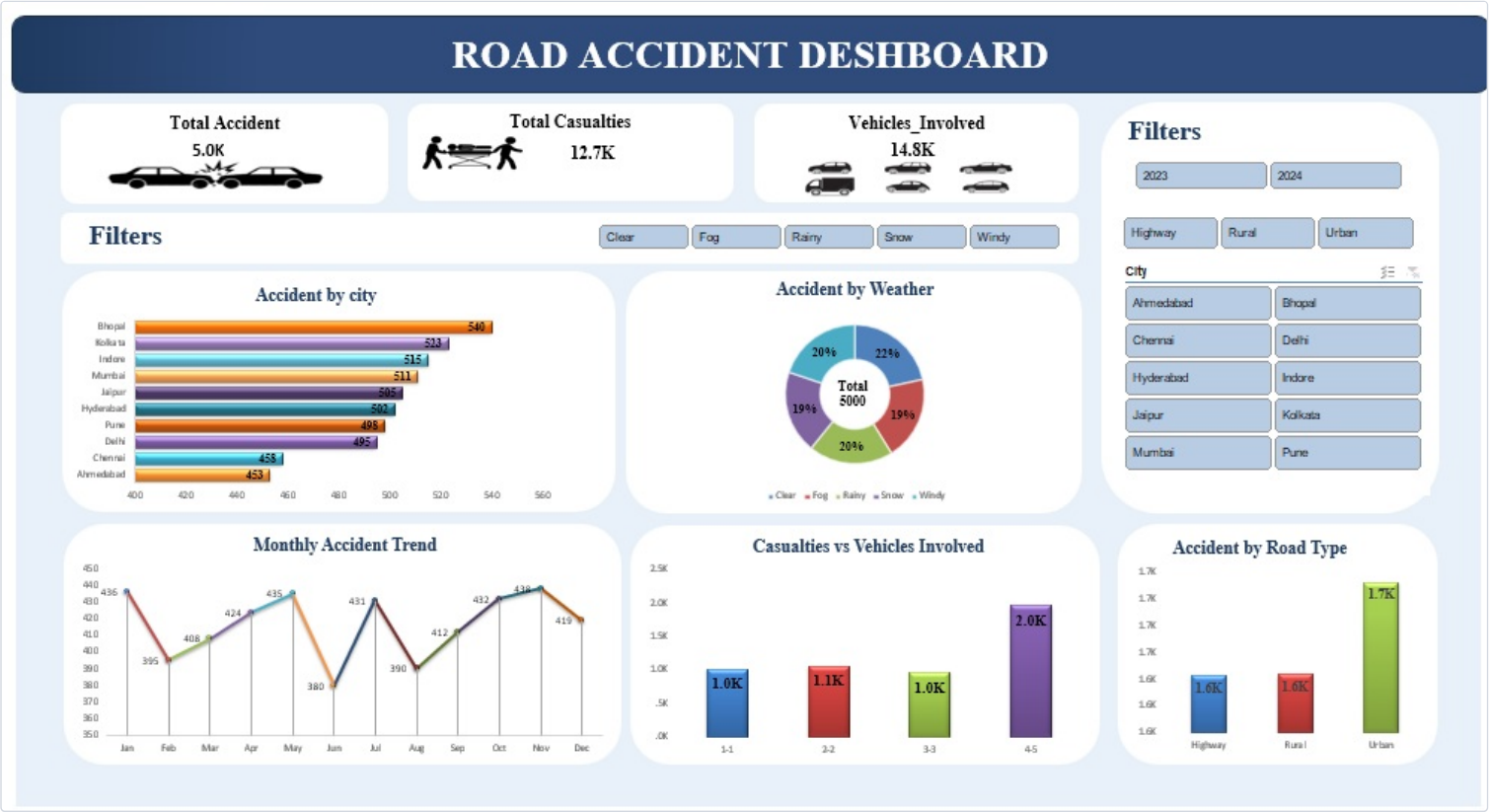
Visualizations

Five charts round out the view: Accident by City, Accident by Weather, Monthly Accident Trend, Casualties vs. Vehicles Involved, and Accident by Road Type.

4.Walkthrough of the Dashboard Views

The clearest way to understand what this dashboard can do is to watch it in action — so the following pages walk through five real filter combinations, each showing how the same underlying data tells a different story depending on the lens applied.

4.1 — Full Dataset, No Filters Applied



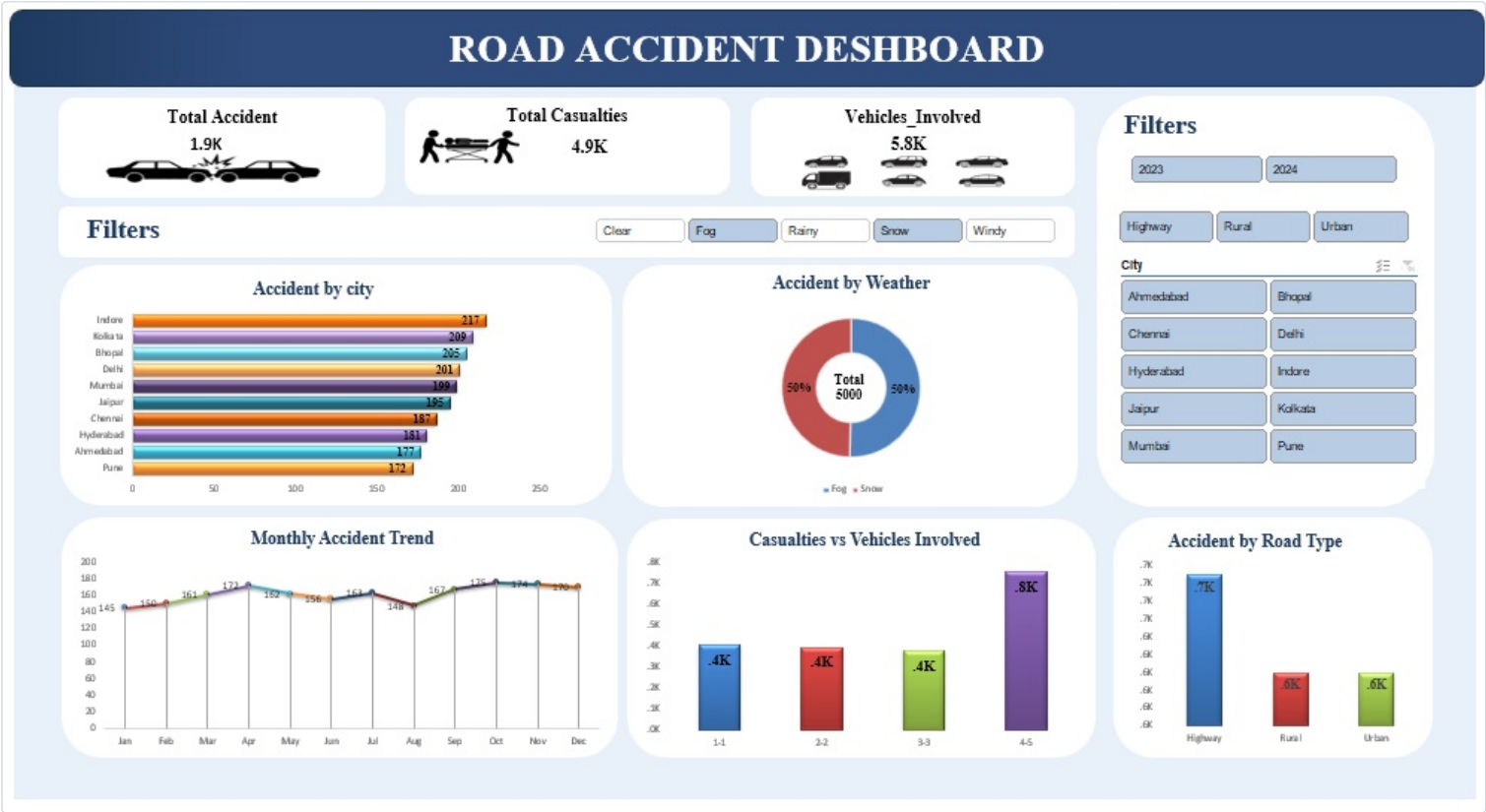
The unfiltered master view — every accident, every city, every condition, all at once.



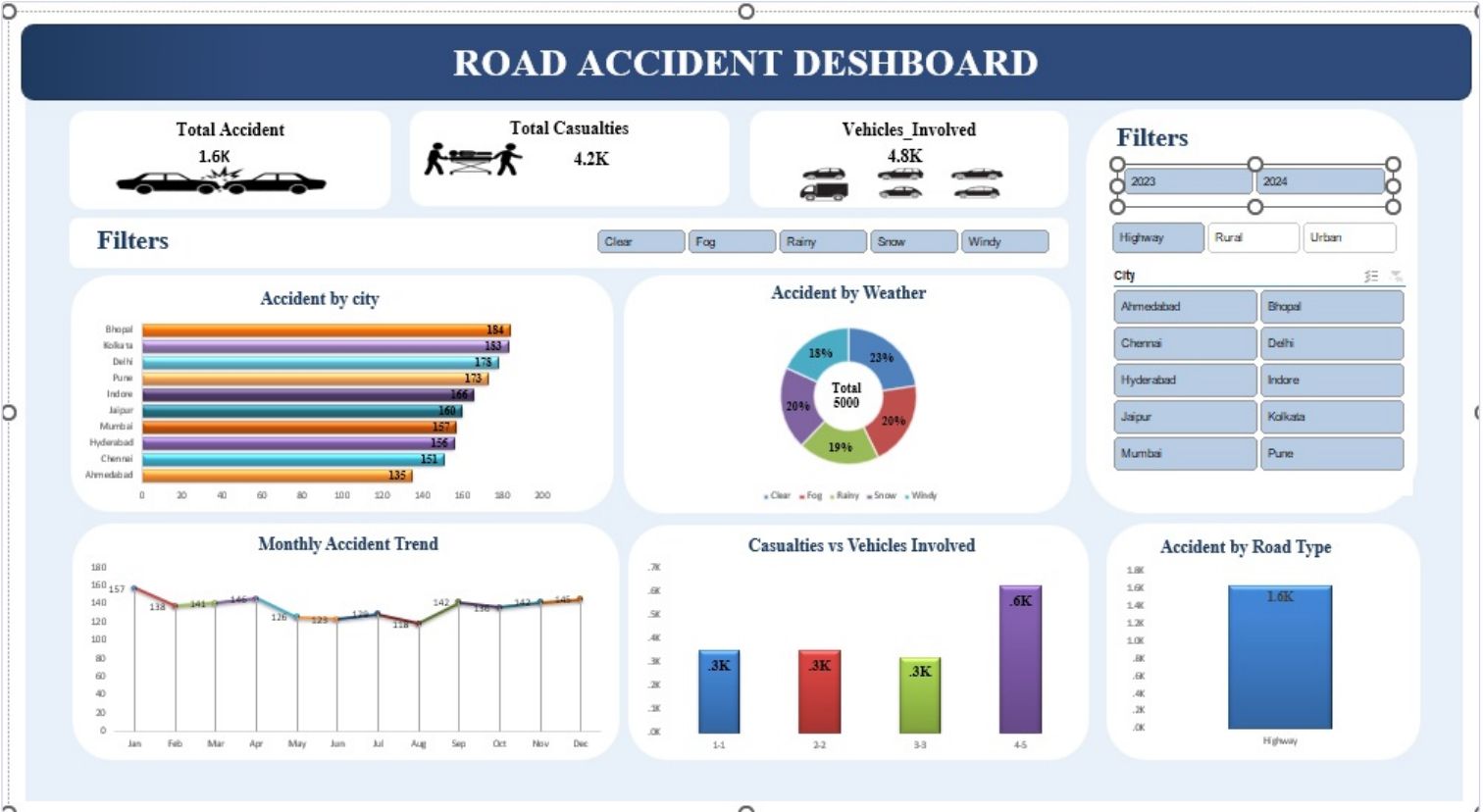
This is the dashboard at rest, showing the complete five-thousand-record dataset. On average, each accident involves close to 2.5 casualties and nearly 3 vehicles — a reminder that these aren't minor fender-benders but multi-party, multi-injury events.

Looking city by city, **Bhopal** sits at the top with 540 accidents, just ahead of Kolkata (533), Indore (515), and Mumbai (511); Ahmedabad reports the fewest at 453. The weather breakdown is fairly balanced, with Clear conditions accounting for the largest single slice at 22% — a useful reminder that "bad weather" isn't the whole story behind accident frequency. Month to month, totals drift between roughly 380 and 440, dipping in June and peaking in April and November, which points to mild seasonal rhythm rather than one dramatic high-risk window. Finally, **Urban roads** edge out Highway and Rural roads as the single largest contributor to accident volume, at 1.7K versus 1.6K each.

4.2 — Filtered by Weather: Fog & Snow



4.3 — Filtered by Road Type: Highway

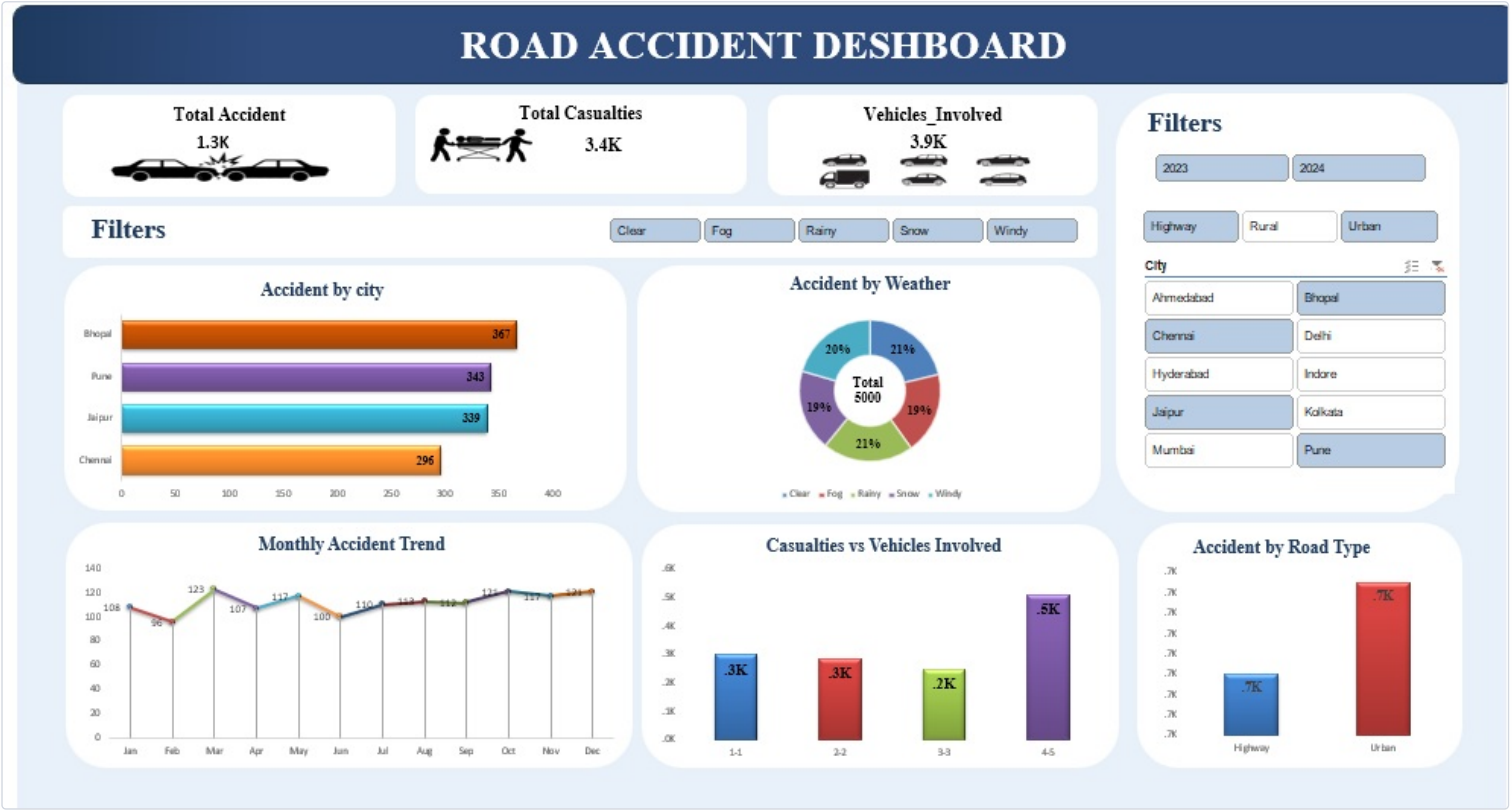


Looking exclusively at highway accidents.

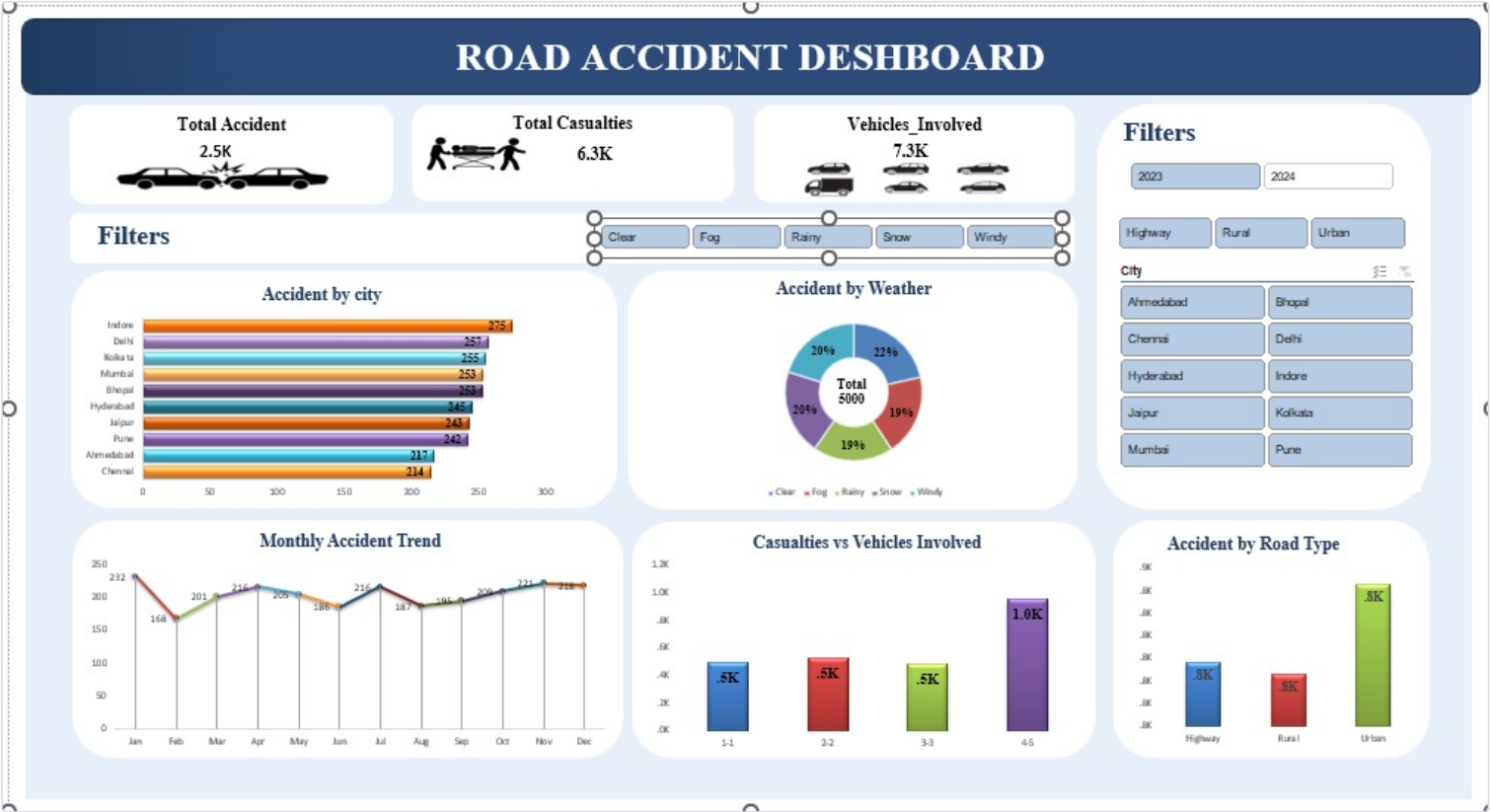


Highway accidents alone total 1.6K, with Bhopal once again leading at 184, followed by Kolkata (183) and Delhi (178). One detail stands out on closer inspection: the casualty-to-accident ratio here runs slightly higher than the dataset-wide average (roughly 2.6 versus 2.5), a small but plausible signal that highway accidents tend to be a touch more severe — likely tied to higher travel speeds. Fog (23%) and Rainy conditions (20%) are the most common weather backdrop for these incidents, reinforcing that visibility and road grip matter most where speeds are highest.

4.4 — Filtered by City: Bhopal, Pune, Jaipur & Chennai



4.5 — Filtered by Year: 2023, with Fog & Snow



Combining two slicers at once — Year and Weather — to test for compounding effects.



This view stacks two filters together — Year 2023 and Fog/Snow weather — and the result is a useful illustration of why multi-slicer analysis matters. The city leaderboard flips again: **Indore** tops the list at 275 accidents, ahead of Delhi (257) and Mumbai (255) — neither of which led in the unfiltered view. On its own, neither the Year filter nor the Weather filter would have surfaced this; it only becomes visible once they're combined. That's the real value of an interactive dashboard over a static report: it lets the analyst ask compound questions on the fly, rather than being limited to whatever cross-tabs were pre-built into a PDF.

5.Chart Reference Guide

For quick reference, here's what each visualization on the dashboard is built to answer:

Chart	What it answers
Accident by City	Which cities see the most accidents, and by how much
Accident by Weather	How accident share splits across Clear, Fog, Rainy, Snow, and Windy conditions
Monthly Accident Trend	Whether accidents cluster in certain months or stay roughly flat year-round
Casualties vs. Vehicles Involved	How accident severity scales with the number of people and vehicles involved
Accident by Road Type	Whether Highway, Rural, or Urban roads carry the greatest share of risk

6.Key Insights & Recommendations

Pulling everything together, a few findings come through clearly and consistently across the different filtered views:

- **Urban roads carry the heaviest overall burden** — even though Highway accidents trend slightly more severe per incident, sheer volume puts Urban roads at the top of the list.
- **Bhopal is the dashboard's most persistent hotspot**, appearing at or near the top of nearly every filtered view, regardless of weather or year. This is a strong candidate for targeted intervention.
- **Weather changes the leaderboard, not just the totals**. Cities like Indore rise sharply under fog and snow conditions even though they don't lead the unfiltered rankings — suggesting infrastructure or driving conditions there are more weather-sensitive than elsewhere.
- **Seasonality is mild but real**, with a consistent dip in June and peaks in spring and late autumn — useful for timing awareness campaigns or enforcement pushes.
- **Severity tracks loosely with road type and speed**, with highway accidents showing a slightly higher casualty-to-accident ratio than the dataset average.

Based on these patterns, a few practical next steps are worth considering: prioritize road safety audits and infrastructure review in Bhopal and other repeat-offender cities; deploy weather-triggered driver alerts or variable speed limits in cities like Indore during fog and snow events; and time public safety campaigns to land just ahead of the seasonal upticks observed in spring and late autumn.

7.Closing Thoughts

What this project really demonstrates is that Excel, used well, can do far more than store numbers — it can become a genuine decision-support tool. With nothing more than Pivot Tables, Pivot Charts, Slicers, and a bit of careful formatting, a flat dataset of five thousand accident records turns into something an analyst can actually interrogate: filter it, compare it, and trust the answer that comes back.

This dashboard is offered as a working example of that idea — practical, interactive, and built to support real conversations about road safety, not just report on the past.